



**Introduction to
Internet of Things
Assignment-Week 4**

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15

Total marks: $15 \times 1 = 15$

QUESTION 1:

What does the ultrasonic sensor measure generally in the AID system?

- a. Humidity
- b. Distance of the object
- c. Soil pH
- d. Light intensity

Correct Answer: b. Distance of the object

Detailed Solution: In the Agricultural Intrusion Detection system, the ultrasonic sensor measures the distance at which the detected object is located.

See Lecture 16: Sensor Networks – Part III @ 13:22

QUESTION 2:

Which component processes the data received from the sensors in the AID system?

- a. Camera
- b. NFC tag
- c. Microcontroller
- d. Ultrasonic sensor

Correct Answer: c. Microcontroller

Detailed Solution: The microcontroller processes the data received from the sensors in the AID system.

See Lecture 16: Sensor Networks - Part III @ 17:07



QUESTION 3 (MSQ):

How can standard nanodevices communicate?

- a. Electromagnetic communication
- b. Molecular communication
- c. Wired Ethernet
- d. ZigBee communication

Correct Answer: a, b

Detailed Solution: The lecture discusses electromagnetic and molecular communication for nanodevices.

See Lecture 16: Sensor Networks - Part III @ 25:11

QUESTION 4:

When does sensing coverage guarantee network connectivity?

- a. When the transmission range is at least twice the sensing range
- b. When the sensing range is at least twice the transmission range
- c. When the transmission and sensing ranges are equal
- d. When the transmission range is smaller than the sensing range

Correct Answer: a. When the transmission range is at least twice the sensing range

Detailed Solution: Sensing coverage guarantees connectivity when the transmission range is at least twice the sensing range.

See Lecture 17: Sensor Networks - Part IV @ 04:14



QUESTION 5:

Which of the following is a coverage problem in a static WSN?

- a. Area coverage
- b. Data compression
- c. Route encryption
- d. Clock synchronization

Correct Answer: a. Area coverage

Detailed Solution: Area coverage is a standard coverage problem in a static wireless sensor network. The other options are networking tasks, not coverage categories.

See Lecture 17: Sensor Networks - Part IV @ 11:30

QUESTION 6:

Which reporting method is used for forest-fire monitoring in a WSN?

- a. Event-driven reporting
- b. On-demand reporting
- c. Manual reporting
- d. Location-based reporting

Correct Answer: a. Event-driven reporting

Detailed Solution: Forest-fire monitoring uses event-driven reporting because sensor nodes report data when a fire-related event is detected. The lecture contrasts this with on-demand reporting, such as an inventory-control system.

See Lecture 17: Sensor Networks – Part IV @ 01:44.



QUESTION 7:

What does a data mule do?

- a. Replaces all sensor nodes
- b. Measures only temperature
- c. Collects data and carries it to the sink
- d. Repairs damaged hardware

Correct Answer: c. Collects data and carries it to the sink

Detailed Solution: A data mule visits sensor nodes, collects their data, and delivers it to the sink.

See Lecture 18: Sensor Networks - Part V @ 07:15

QUESTION 8:

What may happen if a critical stationary sensor node fails?

- a. The network may be partitioned
- b. All nodes become mobile
- c. The sensing range doubles
- d. The network repairs itself by movement

Correct Answer: a. The network may be partitioned

Detailed Solution: Failure of a critical stationary node may divide the network into disconnected parts.

See Lecture 18: Sensor Networks - Part V @ 00:43



QUESTION 9:

A person who provides sensing information acts as a:

- a. Router
- b. Radio channel
- c. Actuator
- d. Data source

Correct Answer: d. Data source

Detailed Solution: In human-centric sensing, a person may act as a data source by providing or collecting information.

See Lecture 18: Sensor Networks - Part V @ 12:35

QUESTION 10:

Which is an advantage of a stationary wireless sensor network?

- a. Easy topology maintenance
- b. Automatic movement after node failure
- c. Continuously changing topology
- d. Unpredictable sensor locations

Correct Answer: a. Easy topology maintenance

Detailed Solution: In a stationary wireless sensor network, sensor nodes remain fixed. Therefore, deployment and topology maintenance are comparatively easier.

See Lecture 18: Sensor Networks – Part V @ 00:43



QUESTION 11:

State whether the following statement is True or False.

Statement: In a UAV network, the network topology may change frequently because the UAVs are mobile.

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: UAVs move from one location to another, causing frequent changes in their relative positions and communication links. Therefore, the network topology may change frequently.

See Lecture 19: UAV Networks @ 05:23

QUESTION 12:

Which is a common issue in UAV networks?

- a. Frequent topology changes
- b. Fixed relative positions
- c. Permanent communication links
- d. No possibility of UAV failure

Correct Answer: a. Frequent topology changes

Detailed Solution: UAV movement causes frequent changes in network topology and communication links.

See Lecture 19: UAV Networks @ 05:23



QUESTION 13:

Compared with traditional SCADA, M2M supports:

- a. Only isolated systems
- b. No device communication
- c. Continuous human control
- d. Cross-platform integration

Correct Answer: d. Cross-platform integration

Detailed Solution: Traditional SCADA commonly uses isolated proprietary systems, while M2M supports cross-platform integration.

See Lecture 20: Machine to Machine Communication @ 04:28

QUESTION 14:

State whether the following statement is True or False.

Statement: In a stationary wireless sensor network, sensor nodes can automatically move to restore connectivity after a node failure.

- a. True
- b. False

Correct Answer: b. False

Detailed Solution: In a stationary wireless sensor network, sensor nodes remain fixed. Therefore, they cannot move automatically to repair connectivity after a node failure.

See Lecture 18: Sensor Networks – Part V @ 00:43



QUESTION 15:

Which statement best describes M2M communication?

- a. Devices can exchange data without direct human control
- b. Every data exchange must be performed manually
- c. Only one device can communicate at a time
- d. Devices cannot communicate through a network

Correct Answer: a. Devices can exchange data without direct human control

Detailed Solution: M2M communication allows machines or devices to exchange data automatically. Human intervention may still be required for maintenance and operational stability.

See Lecture 20: Machine to Machine Communication @ 09:24.

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